

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5110

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

*I **SAIRAM D 192424111**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- Compute n and $\phi(n)$
- Choose $e=7$ and find the private key d
- Encrypt the message $M=9$
- Decrypt the ciphertext and verify the result
- Analyze how changing e affects security

SOLUTION:

Given:

$p = 11$, $q = 13$, $e = 7$, Message (M) = 9

(a) Compute n and $\phi(n)$

Formula:

$$n = p \times q$$

$$\phi(n) = (p - 1)(q - 1)$$

Calculation:

$$\begin{aligned} n &= 11 \times 13 \\ &= 143 \end{aligned}$$

$$\begin{aligned} \phi(n) &= (11 - 1)(13 - 1) \\ &= 10 \times 12 \\ &= 120 \end{aligned}$$

Answer:

$$\begin{aligned} n &= 143 \\ \phi(n) &= 120 \end{aligned}$$

(b) Find the Private Key (d)

Formula:

$$e \times d \equiv 1 \pmod{120}$$

Given:

$$e = 7$$

Find d such that:

$$7 \times d \equiv 1 \pmod{120}$$

Checking:

$$7 \times 103 = 721$$

$$721 = 120 \times 6 + 1$$

Therefore,

$$d = 103$$

Public Key = (7, 143)

Private Key = (103, 143)

(c) Encrypt the Message

Formula:

$$C = M^e \pmod{n}$$

Substitute:

$$C = 9^7 \pmod{143}$$

Calculation:

$$9^2 = 81$$

$$9^4 = 81^2$$

$$= 6561$$

$$6561 \pmod{143} = 126$$

$$9^7 = 9^4 \times 9^2 \times 9$$

$$= 126 \times 81 \times 9$$

$$126 \times 81 = 10206$$

$$10206 \bmod 143 = 53$$

$$53 \times 9 = 477$$

$$477 \bmod 143 = 48$$

Ciphertext:

$$C = 48$$

(d) Decrypt the Ciphertext

Formula:

$$M = C^d \bmod n$$

Substitute:

$$M = 48^{103} \bmod 143$$

Using modular exponentiation,

$$48^{103} \bmod 143 = 9$$

Recovered Message:

$$M = 9$$

Verification:

Original Message = 9

Decrypted Message = 9

Hence, encryption and decryption are verified successfully.

(e) Effect of Changing e

If e is small (Example: $e = 3$)

- Encryption is faster.
- Less secure if proper padding is not used.
- Vulnerable to low exponent attacks.

If e is large (Example: $e = 65537$)

- Encryption is slightly slower.
- More secure.
- Resistant to many attacks.
- Widely used in modern RSA.

Conclusion:

Small e → Faster but lower security.

Large e → Slightly slower but higher security.

Recommended value:

$$e = 65537$$

Final Answer:

$$n = 143$$

$$\phi(n) = 120$$

Public Key = (7, 143)

Private Key = (103, 143)

Plaintext = 9

Ciphertext = 48

Decrypted Message = 9

Conclusion:

RSA successfully encrypts the plaintext 9 into ciphertext 48 and decrypts it back to 9 using the private key. The original message is recovered successfully, proving that the RSA algorithm works correctly.

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

SOLUTION:

Given:

DES Key Size = 56 bits, Encryption Time = 2 ms/block ,
3-DES Key Size = 168 bits ,Encryption Time = 6 ms/block,
Blowfish Key Size = 128 bits ,Encryption Time = 1
ms/block, Number of Data Blocks = 1000

(i) Calculate Total Encryption Time

Formula:

Total Encryption Time = Number of Blocks $\times$ Time per Block

DES:

$$= 1000 \times 2 \text{ ms}$$

$$= 2000 \text{ ms}$$

$$= 2 \text{ seconds}$$

3-DES:

$$= 1000 \times 6 \text{ ms}$$

$$= 6000 \text{ ms}$$

= 6 seconds

Blowfish:

= 1000×1 ms

= 1000 ms

= 1 second

Total Encryption Time Table

Algorithm Time per Block Total Time

DES 2 ms 2000 ms (2 s)

3-DES 6 ms 6000 ms (6 s)

Blowfish 1 ms 1000 ms (1 s)

(ii) Analyze Trade-offs Between Security and Performance

DES:

- Key Size = 56 bits
- Fast encryption
- Very weak security
- Easily broken by brute-force attacks
- Not recommended for modern applications

3-DES:

- Key Size = 168 bits
- Performs DES encryption three times
- Provides much stronger security than DES
- Slowest among the three algorithms
- Suitable where security is more important than speed

Blowfish:

- Key Size = 128 bits
- Strong security
- Fastest encryption speed
- Efficient for both software and hardware implementations
- Suitable for real-time applications

(iii) Recommend the Best Algorithm for Secure Real-Time Communication

For secure real-time communication, Blowfish is the best choice

because:

- It has the fastest encryption time (1 second for 1000 blocks).

- It provides strong security with a 128-bit key.
- It requires less processing time than DES and 3-DES.
- It is suitable for systems where both speed and security are

important. **Final Answer:**

DES Encryption Time = 2000 ms (2 seconds)

3-DES Encryption Time = 6000 ms (6 seconds)

Blowfish Encryption Time = 1000 ms (1 second)

Best Algorithm:
Blowfish

Conclusion:

Among DES, 3-DES, and Blowfish, Blowfish offers the best balance between security and performance. It provides strong encryption while requiring the least encryption time, making it the most suitable algorithm for secure real-time communication.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
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Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation